

FixHeatmap v2.2 - Edge Score with Decay & Calibration

Business Background

The purpose of this module is to support speculative trading decisions between Fixing1 and Fixing2. The existing heatmap shows historical direction of profitable arbitrage opportunities. However, directional frequency alone is insufficient because it does not consider the economic value of the opportunity nor the fact that recent observations are usually more relevant than older observations. The Edge Score with Decay module combines: directional probability (LONG / SHORT) magnitude of historical spreads time decay (recent observations have higher importance) The objective is to identify hours where historical behaviour suggests both a high probability and a meaningful economic edge.

1. Functional Objective

Create a new dashboard component called: EDGE SCORE WITH DECAY The module shall calculate a directional trading signal for each hour H01-H24 and rank hours according to the strength of the expected edge.

2. User Parameters

The user shall be able to modify: Lookback Window: 3 / 5 / 7 / 14 / 30 days Decay Lambda: 0.6 / 0.7 / 0.8 / 0.9 / 1.0 Recommended trading settings: Lookback = 5 days Lambda = 0.8

3. Decay Logic

Recent observations shall have greater importance than older observations. Weight calculation: $\text{weight} = \lambda^{(N-1-k)}$ where: N = lookback window length k = observation index λ = decay parameter The newest observation receives weight 1.0. Older observations receive exponentially smaller weights.

4. Directional Probability

For every hour H01-H24: spread = Fixing2 - Fixing1 Positive spread: Buy F1 / LONG Negative spread: Sell F1 / SHORT Weighted probability: $P_{\text{buy_weighted}} = \frac{\text{SUM}(\text{weight} \times \text{is_buy})}{\text{SUM}(\text{weight})}$ $P_{\text{sell_weighted}} = 1 - P_{\text{buy_weighted}}$

5. Weighted Direction

$\text{weighted_direction} = 2 \times P_{\text{buy_weighted}} - 1$ Interpretation: +1 = full LONG dominance 0 = no directional edge -1 = full SHORT dominance

6. Edge Score Calculation

For each hour: Edge Score = $\text{weighted_direction} \times P75(\text{abs}(\text{spread}))$ P75 shall be calculated for the selected lookback window and hour. Purpose: Combine probability of success with economic value of the spread.

7. Dashboard Visualisation

New chart: EDGE SCORE WITH DECAY Requirements: X axis: Hours H01-H24 Y axis: Edge Score Green bars: LONG edge Red bars: SHORT edge Bar height: Absolute value of Edge Score

8. Tooltip

Each bar tooltip shall display: Hour Lookback Window Lambda Weighted Probability Unweighted Probability P75 Absolute Spread Edge Score Suggested Direction

9. Hour Ranking Table

Generate ranking table sorted by absolute Edge Score. Columns: Hour Direction Weighted Probability Unweighted Probability P75 Spread Edge Score Signal Strength Suggested Action

10. Edge Calibration Module

Add a separate analytical module: EDGE CALIBRATION Purpose: Verify whether larger Edge values historically produced higher trading profitability. For each Edge bucket calculate: Number of Observations Hit Ratio Average PnL Median PnL Total PnL

11. Trading Interpretation

The calibration module shall automatically determine optimal Edge thresholds. Thresholds must not be hardcoded. The system should derive thresholds from historical backtesting results and assign categories: No Edge Weak Signal Moderate Signal Strong Signal Exceptional Signal This allows the platform to adapt automatically when market behaviour changes.

12. Expected Business Outcome

The trader should be able to identify: - hours with the strongest historical edge, - hours where LONG is preferred, - hours where SHORT is preferred, - whether the edge is strengthening or weakening, - whether the signal is statistically meaningful and economically attractive. The module is intended as a decision-support tool and not as an automatic trading system.